Linear systems and matrix notation

Linear equation: $a_1x_1 + \cdots + a_nx_n = b$; real scalars a_i are **coefficients**, b is the **constant**, and x_i are **unknowns**.

Linear system: m linear equations in n unknowns, written

$$a_{i1}x_1 + \cdots + a_{in}x_n = b_i \quad (1 \le i \le m), \qquad A\mathbf{x} = \mathbf{b}.$$

Matrices and vectors: coefficient matrix $A = (a_{ij}) \in \mathbb{R}^{m\times n}$; unknown vector $\mathbf{x} \in \mathbb{R}^n$; constant vector $\mathbf{b} \in \mathbb{R}^m$; augmented matrix $[A \mid \mathbf{b}] \in \mathbb{R}^{m\times(n+1)}$.

Solution: values for all n unknowns satisfying every equation.

Consistent / inconsistent: at least one solution / no solution.

Equivalent systems: exactly the same solution set.

Planes and lines in $\mathbb{R}^3$

Plane: through $\mathbf{x}_0 = (x_0, y_0, z_0)$ with nonzero **normal vector** $\mathbf{n} = (n_1, n_2, n_3)$:

$$\mathbf{n} \cdot (\mathbf{x} - \mathbf{x}_0) = 0 \quad \Longleftrightarrow \quad n_1(x - x_0) + n_2(y - y_0) + n_3(z - z_0) = 0.$$

General plane equation: $ax + by + cz = d$, where $(a, b, c) \neq (0, 0, 0)$ and $d = ax_0 + by_0 + cz_0$ for a point on the plane.

Parametric line: point $\mathbf{x}_0$, nonzero **direction vector** $\mathbf{d} = (d_1, d_2, d_3)$, $t \in \mathbb{R}$:

$$\mathbf{x} = \mathbf{x}_0 + t\mathbf{d}.$$

Symmetric form: if all $d_i \neq 0$,

$$\frac{x - x_0}{d_1} = \frac{y - y_0}{d_2} = \frac{z - z_0}{d_3} = t.$$

Line as intersection: two planes through $\mathbf{x}_0$ with nonparallel normal vectors $\mathbf{n}_1, \mathbf{n}_2$ meet in the line defined by $\mathbf{n}_1 \cdot (\mathbf{x} - \mathbf{x}_0) = 0$ and $\mathbf{n}_2 \cdot (\mathbf{x} - \mathbf{x}_0) = 0$.

Row operations and equivalence

Elementary row operations: interchange $R_i \leftrightarrow R_j$; scale $R_i \to cR_i$ with $c \neq 0$; replace $R_i \to R_i + cR_j$ with $i \neq j$ and $c \in \mathbb{R}$. Rows not replaced stay unchanged.

Row equivalent: same-size matrices connected by a finite sequence of elementary row operations.

Preservation theorem: row-equivalent augmented matrices for two systems of m equations in n unknowns give equivalent systems. The same operations on equations preserve solutions.

Echelon forms and pivots

Row echelon form (REF): zero rows are at the bottom; the leftmost nonzero entry in each nonzero row is 1 (its **leading one**); leading ones move strictly right as rows descend.

Pivot entry / position / column: in REF, a leading one / its location / a column containing one.

Reduced row echelon form (RREF): REF with every pivot column zero except at its leading one.

Uniqueness theorem: every matrix is row equivalent to a unique RREF.

Gaussian elimination and variables

Algorithm: The Gaussian elimination algorithm. In the remaining rows, take the leftmost column with a nonzero entry a; move its row to the top of those rows, scale by $1/a$, then subtract multiples of that row to clear entries below its leading one. Repeat on the rows below. Stop if no rows remain or all remaining rows are zero. The result is a row-equivalent REF matrix.

Echelon system: a linear system whose augmented matrix is in REF.

Leading (basic) variables: variables in pivot columns of an REF augmented matrix. Only columns *left of the vertical line* correspond to variables.

Free variables: variables in the non-pivot variable columns.

Algorithm: Solving $A\mathbf{x} = \mathbf{b}$ by Gaussian elimination. Form $[A \mid \mathbf{b}]$ and reduce it to REF. Check for inconsistency; if consistent, choose the free variables as arbitrary real parameters. Use **back substitution**, from the bottom nonzero row upward, to express the basic variables in terms of those parameters.

Consistency and number of solutions

Inconsistency test: no solution exactly when an REF augmented matrix has a row

$$[\,0 \quad \cdots \quad 0 \mid c\,], \qquad c \neq 0.$$

Equivalently, there is a pivot in the constant column, right of the vertical line.

Parameter-count theorem: for a *consistent* system with n variables and r pivot columns left of the vertical line in REF, the solution set has exactly $n - r$ free parameters.

Unique solution: consistent and $r = n$ (every variable column is a pivot column).

Infinitely many solutions: consistent and $r < n$ (at least one free variable).

Homogeneous systems and basic solutions

Homogeneous system: $A\mathbf{x} = \mathbf{0}$, with $A \in \mathbb{R}^{m \times n}$. It is always consistent: $\mathbf{x} = \mathbf{0}$ is the **trivial solution**. Any nonzero solution is **nontrivial**.

Existence: if $n > m$, a homogeneous system has a nontrivial solution.

Closure under linear combinations: if $A\mathbf{u}_i = \mathbf{0}$ for $1 \leq i \leq k$, then $A(c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k) = \mathbf{0}$ for all $c_1, \ldots, c_k \in \mathbb{R}$.

Basic solutions: for each free variable in an REF homogeneous system, set it to 1, all other free variables to 0, and solve for the basic variables.

Basic-solution theorem: if r is the number of pivots in an REF of A, there are exactly $n - r$ basic solutions, one per free variable. Every solution has a unique expression

$$\mathbf{x} = t_1\mathbf{v}_1 + \cdots + t_{n-r}\mathbf{v}_{n-r}, \qquad t_1, \ldots, t_{n-r} \in \mathbb{R},$$

where the $\mathbf{v}_i$ are those basic solutions. If $r = n$, there are no free variables or basic solutions; the only solution is $\mathbf{0}$.